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Chaotic Amplitude Control for the Ising Minimization using Optical Parametric Oscillator Systems

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Abstract

This thesis investigates performance challenges and optimization techniques for coherent Ising machines (CIMs), a class of quantum-optical devices designed to solve combinatorial optimization problems by mapping them onto the ground state of the Ising model. We examine two key issues affecting CIM performance: the freeze-out phenomenon, which can trap the system in local minima, and amplitude heterogeneity. To address these challenges, we analyze three novel error correction schemes: CIM with chaotic amplitude control (CIM-CAC), CIM with chaotic feedback control (CIM-CFC), and CIM with separable feedback control (CIM-SFC). Through numerical simulations, we compare the dynamics, correlation properties, and trajectories of these systems. We evaluate their effectiveness in solving MAX-CUT problems on various graph types using G-Set benchmark instances. Additionally, we explore energy reduction strategies for optical implementations of CIMs using pump pulse schemes with second harmonic generation. Our findings provide insights into the strengths and limitations of each approach in mitigating performance issues and escaping local minima.

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1 Introduction

Combinatorial optimization problems, often classified as NP-hard, are widely encountered in fields such as drug discovery, finance, circuit design, and manufacturing [McMahon et al., 2016]. Despite significant advancements in algorithms and digital computing technology, the traditional von Neumann architecture faces substantial costs to achieve marginal performance gains when dealing with the complexities of large-scale optimization problems [Stroev and Berloff, 2023]. This challenge has prompted a notable shift towards adopting more sustainable computational paradigms [Thompson and Spanuth, 2021]. Specifically, two main strategies have emerged [Cummins and Berloff, 2024]: improving algorithms within current computational frameworks, and exploring novel hardware paradigms rooted in physics principles including entropy minimization, enhanced energy efficiency, and reduced dissipation [Vadlamani et al., 2020], as well as quantum phenomena like superposition and entanglement [Farhi et al., 2000, Kadowaki and Nishimori, 1998, Brooke et al., 2001]. While the former approach demonstrates minor advancements, the latter aims to tackle complex optimization challenges beyond current methods. It does this by integrating analog, physics-based algorithms and hardware [Cummins and Berloff, 2024], mapping the cost function to the energy landscape of simple physical systems, such as the classical Ising [Lucas, 2014] and XY models [Berloff et al., 2017, Kalinin et al., 2020], to find the ground state solution efficiently.

Lasers are promising open dissipative systems that exhibit second-order phase transitions at the oscillation threshold. In a laser, potentially oscillating modes compete for available gain, reducing the gain accessible to other modes due to the cross-saturation effect [Siegman, 1986]. The mode with the minimum threshold gain is most likely to be excited first, giving it an edge over other modes. This phenomenon is known as the minimum gain principle [Wang et al., 2013]. To engineer the photon decay rate in a mutually injection-locked laser network to form an Ising Hamiltonian [Utsunomiya et al., 2011], each laser is made polarization-degenerate. It represents an Ising spin of +1 when right circularly polarized photons dominate and -1 when left circularly polarized photons dominate. The collective polarization states of all lasers form the global modes of the network. Numerical simulations support the minimum gain principle, showing that the network oscillates in global modes with the minimum photon decay rate. Additionally, the transient time to reach steady state is determined by mutual injection signals among the lasers and does not scale with the number of Ising spins [Takata et al., 2012].

Degenerate optical parametric oscillators (DOPOs) also function as open dissipative systems that exhibit second-order phase transitions at the oscillation threshold [Woo and Landauer, 1971]. Above this threshold, due to phase-sensitive amplification, DOPOs can only oscillate in one of two discrete phases: 0 or π relative to the pump laser. This bistability was previously exploited in early digital computers for logic circuits [Goto, 1959]. In DOPOs, quantum noise during optical parametric down-conversion determines the phase [Louisell et al., 1961]. This characteristic was utilized to create a quantum random-number generator by XORing the phases of two independently oscillating OPOs [Marandi et al., 2012].

A notable recent development in the field of combinatorial optimization is the construction of DOPO-based optical artificial spin networks using photonics technologies. These networks map spins onto the bistability of their optical phase. One such quantum-optical device is the Coherent Ising Machine (CIM). The CIM is designed to address combinatorial optimization problems by mapping these instances onto ground-state search problems of the Ising model using polynomial resources. By implementing the Ising model in physical systems rather than conventional digital computers, the CIM enables a natural evolution towards the Ising ground state [McMahon et al., 2016].

In this thesis, we aim to provide a comprehensive summary and clarification of the operational principles of CIMs. We will demonstrate how CIMs harness the coherent nature of light to perform computations, highlighting their potential for parallel processing and high speed, which makes them suitable for tackling specific complex computational problems that are intractable for classical computers. Additionally, we will investigate performance challenges in CIMs, specifically the freeze-out phenomenon and amplitude heterogeneity, exploring the reasons behind these issues and proposing methodologies to mitigate them. Furthermore, we will compare three novel methodologies: CIM

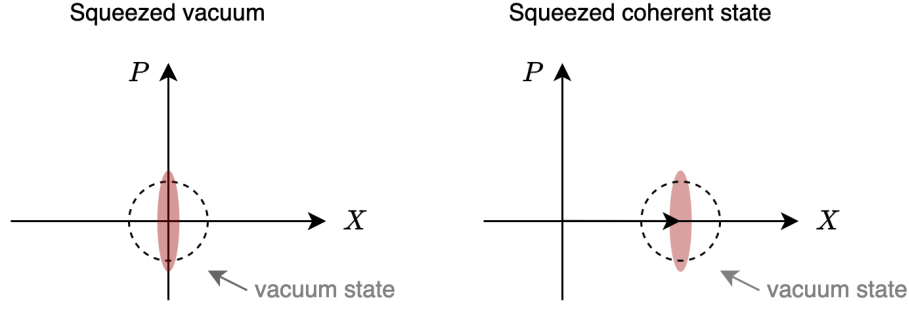


Figure 1: Left: Vacuum state and squeezed vacuum state. Right: Coherent state and squeezed coherent state.

with chaotic amplitude control, CIM with chaotic feedback control, and CIM with separable feedback control. By analyzing their similarities and differences, we aim to provide a deeper understanding of these approaches. To evaluate the effectiveness of these methodologies, we will test their performance on the MAX-CUT problem using G-Set instances of varying graph types, including random, planar, and toroidal graphs. This comparative analysis will provide insights into the strengths and limitations of each approach in solving complex combinatorial optimization problems.

By doing so, we aim to contribute to the development of more efficient and effective computational paradigms for solving NP-hard problems, advancing the field of quantum-optical computing and its applications in real-world scenarios. In alignment with the principles of open science, we further aim to reproduce and verify the claim in Leleu et al. (2019) that analogue bistable systems, such as the system of optical oscillators, can find low-energy states of the Ising Hamiltonian at least as efficiently as current state-of-the-art heuristics, even when simulated on a classical computer.

2 Mathematical Framework of the Coherent Ising Machine

This section offers essential mathematical background necessary for comprehending the governing equations of the CIM and its behavior. To understand the underlying principles of the CIM, we employ a semi-classical deterministic framework that approximates the truncated Wigner stochastic differential equation (W-SDE) for a hypothetical quantum-optic CIM [Kako et al., 2020], focusing on scenarios where the deamplification factor G is large ($G \ll 1$) [Wang et al., 2013]:

At the initial time $t = 0$, each signal pulse field is prepared in a vacuum state, and each error pulse field is prepared in a weak coherent state (Figure 1). Upon activation of the pump fields p and p_i at $t \geq 0$, a vacuum field incident on the extraction beamsplitter (BS_e) from an open port is squeezed/anti-squeezed by a phase-sensitive amplifier (PSA) in this optical delay line (ODL) CIM. Specifically, the vacuum fluctuation in the in-phase component $\tilde{X} = \frac{1}{2}(\hat{a} + \hat{a}^\dagger)$ is deamplified by a factor of $1/G$, while the vacuum fluctuations in the quadrature-phase component $\tilde{P} = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger)$ are amplified by a factor of G . Similarly, the vacuum fluctuations incident on the OPO pulse field due to any linear loss of the cavity are all squeezed by the respective PSA. Furthermore, the pump field and feedback injection field fluctuations along the in-phase component are also deamplified by the respective PSA [Reifenstein et al., 2021].

Such CIM architecture employs a linear feedback mechanism to control the time evolution of the signal pulse amplitudes. This is represented by a network of analog bistable units $\vec{x} = \{x_i\}_{i=1}^N$ with $x_i \in \mathbb{R}$:

$$\frac{dx_i}{dt} = \phi(x_i) + z_i, \quad (2.1)$$

where $\phi(x_i)$ describes the time evolution of the isolated units and is defined as $\phi(x_i) = -\frac{\partial V_b}{\partial x_i}$. The paradigmatic bistable potential V_b is given by $V_b = (1-p)\left(\frac{x_i^2}{2}\right) + \left(\frac{x_i^4}{4}\right)$. The function $\phi(x_i)$ can be

expressed as:

$$\phi(x_i) = -x_i + px_i - x_i^3, \quad (2.2)$$

where the terms represent loss, linear gain with rate p , and nonlinear saturation, respectively.

To solve a combinatorial optimization problem defined by the cost function $V(\vec{\sigma})$, the coupling term z_i is chosen as $z_i \propto -\frac{\partial V(\vec{\sigma})}{\partial x_i}$, making $\vec{z}$ the gradient of the potential V . Specifically, for Ising problems with the cost function $V(\vec{\sigma}) = \mathcal{H} = -\frac{1}{2} \sum_{ij} J_{ij} \sigma_i \sigma_j$, where $\mathbf{J}$ is the Ising coupling matrix with elements J_{ij} and $\sigma_i = \text{sign}(x_i)$ represents a potential spin configuration solution, the injection term is given by:

$$z_i = \xi \sum_j J_{ij} x_j = \xi \{ \mathbf{J} \vec{x} \}_i, \quad (2.3)$$

with coupling strength ξ acting as a normalization constant for J_{ij} ($\xi > 0$, $J_{ii} = 0$, and $J_{ij} = J_{ji}$ for $j \neq i$) [Leleu et al., 2019]. Although noise is neglected in this model, the initial amplitudes can be randomly chosen to generate a diverse set of trajectories and explore possible solutions [Reifenstein et al., 2021].

3 Performance Challenges in Coherent Ising Machines

This section addresses performance challenges in coherent Ising machines, focusing on the intrinsic freeze-out phenomenon, which causes trapping in local minima, and amplitude heterogeneity. We discuss the occurrence and implications of freeze-out and analyze its underlying causes by comparing the evolution of CIM DOPOs with the Metropolis-Hastings algorithm. Additionally, we explore the problem of amplitude heterogeneity and offer a comprehensive theoretical introduction to innovative methods designed to mitigate this issue and destabilize local minima in analog spin systems.

3.1 Freeze-out Phenomenon in CIMs

Theoretically, coherent Ising machines (CIMs) are expected to reach the ground state given sufficient computation time for regular lattices [Hamerly et al., 2016]. However, [Böhm et al., 2018] discovered that domain evolution often freezes at a certain point, beyond which the amplitude of each degenerate optical parametric oscillator (DOPO) remains unchanged, and the domain structure becomes fixed. This section examines the dynamics of DOPOs and the impact of operational parameters on CIM performance to understand the freeze-out phenomenon and devise strategies to optimize performance.

Implications of Freeze-Out

We investigate freeze-out phenomena within a 50×50 square lattice where spins interact with anti-ferromagnetic coupling ($J_{ij} = -1$). This lattice type is chosen due to its well-understood properties with an exact analytical solution [Onsager, 1944]: unlike ferromagnetic lattices where all spins align uniformly, the antiferromagnetic lattice displays a checkerboard pattern where adjacent spins alternate between up and down configurations at ground state. This distinction is fundamental as it influences how domain dynamics and freeze-out are observed in our simulations.

To model the system, we follow [Böhm et al., 2018] and incorporate a checkerboard pattern into the DOPO amplitudes to visualize domain dynamics without influencing measurement outcomes. Our study involves varying the normalized pump amplitudes ($p = 0.2, p = 0.5$, and $p = 0.8$), starting from an initial random spin state. This state mirrors the conditions of DOPOs operating in a squeezed vacuum state below their threshold [Yamamoto and Semba, 2016].

Figure 2 illustrates the time evolution of DOPOs during the ground state search on a regular 2D lattice using different pump amplitudes ($p = 0.8, 0.5, 0.2$). Our study reveals a distinct two-phase process in this search: growth and saturation [Hamerly et al., 2016]. Initially, in the growth phase, DOPO amplitudes increase, organizing artificial spins into well-defined domain structures delineated by domain walls (see $Z = 40$). Subsequently, during the saturation phase, as pump energy decreases, DOPOs achieve maximum amplitude, resulting in inward radial movement of domain walls [Böhm

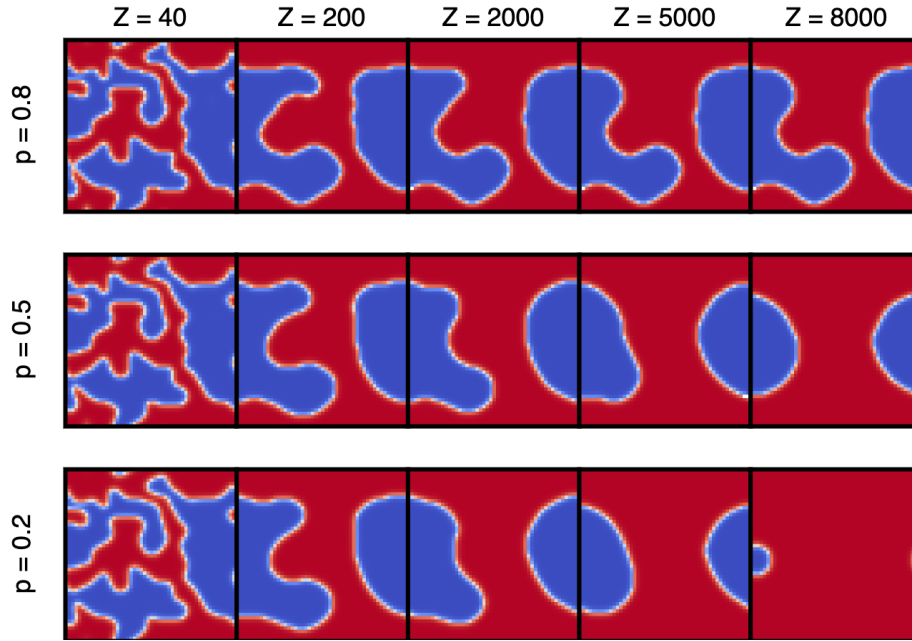


Figure 2: Time evolution of the DOPOs during ground state search for the regular 2D lattice. Snapshots of the time evolution of the x_i 's at different pump amplitudes $p = \{0.8, 0.5, 0.2\}$ visualized in the regular lattice structure. Blue indicated the spin-down and red the spin-up.

et al., 2018]. This evolution leads over time to the shrinkage and eventual disappearance of domains (see for $p = 0.2$, $Z = 200 \rightarrow 8000$).

We observe that the speed of domain wall movement is influenced by wall curvature, where more curved walls move faster [Hamerly et al., 2016]. Notably, at low pump amplitude ($p = 0.2$), the process of domain wall disappearance continues until the ground state is achieved. This validates the effectiveness of CIMs in performing ground state searches for square lattice problems involving a large number of artificial spins.

Additionally, at high pump amplitude ($p = 0.8$), we observe a phenomenon where the evolution of spin amplitudes freezes (see for $p = 0.8$, $Z = 5000 \rightarrow 8000$), indicating that the system becomes trapped in a local minimum spin configuration, known as the freeze-out issue. In such cases, the computational approach fails to escape from this configuration, posing a challenge for achieving the true ground state. Furthermore, we find that higher pump amplitudes accelerate the formation of numerous small domains and domain walls during the initial growth phase. In contrast, lower pump amplitudes near the operational threshold prolong the saturation process, resulting in larger domains that closely resemble the system's ground state.

In Figure 3, we analyze the time evolution of DOPO amplitudes at pump amplitudes $p = 0.2$ and $p = 0.8$. At $p = 0.2$, the system reaches the ground state configuration in approximately 10,000 iterations, whereas at $p = 0.8$, we observe a freeze-out point where the system becomes trapped in local minima and cannot escape. This highlights the critical role of pump amplitude in shaping the behavior of domains. Moreover, we notice that at high pump amplitudes, energy rapidly decreases until freeze-out, abruptly halting further evolution, whereas lower pump intensities near threshold exhibit a more gradual energy decline that closely aligns with the ground state configuration. This phenomenon is due to a higher feedback-to-DOPO amplitude ratio near threshold, which enhances the likelihood of spin state changes during saturation, and intricately linked to the time required for DOPOs to reach saturation or flip amplitudes [Böhm et al., 2018].

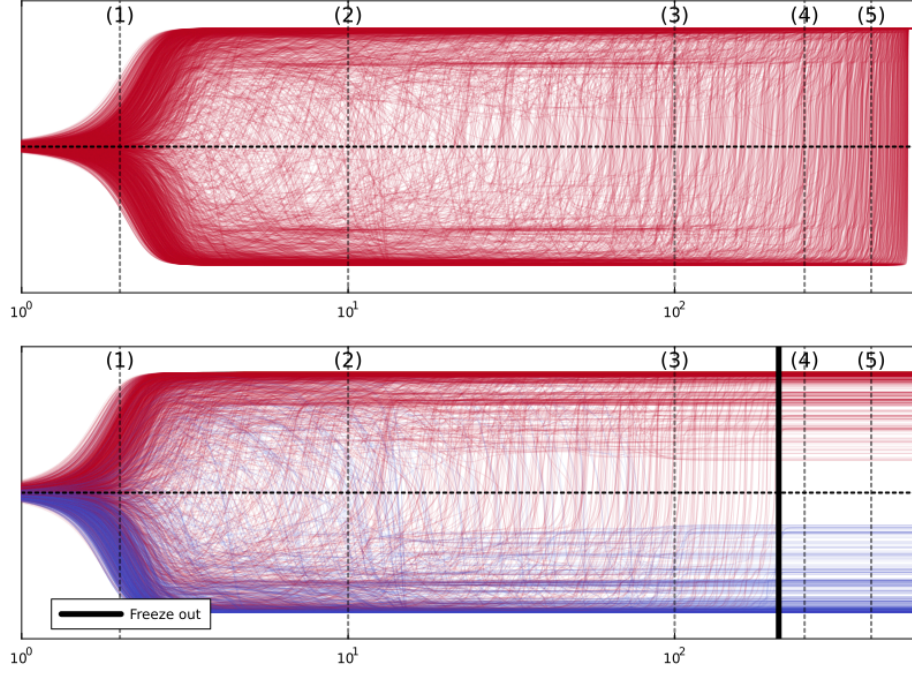


Figure 3: Exemplary time evolution of the CIM. The top plot shows the time evolution of DOPO amplitudes with $p = 0.2$. We can see that the solution reaches the ground state, and does not get trapped in a minima beforehand. The bottom plot shows the time evolution of DOPO amplitudes with $p = 0.8$. We can clearly identify the time point at which the configuration gets trapped in a local minima.

Causes of Freeze-Out

To analyze the freeze-out process, we consider two effects common in Monte Carlo methods: (i) trapping in local energy minima, and (ii) high spin temperatures. Local energy minima are spin configurations difficult to escape once reached, typically associated with parallel domain walls in the 2D square lattice. However, observations of frozen states do not support this, ruling out local energy minima as a cause of freeze-out [Böhm et al., 2018]. High temperatures, on the other hand, lead spins to an equilibrium state far from the actual ground state.

In quantum annealing hardware, non-equilibrium dynamics during annealing are frequently associated with an effective temperature, requiring long annealing times to reach a low effective temperature [Benedetti et al., 2016]. To understand the freeze-out process in CIMs in terms of effective temperature, we compare ground state searches to Markov Chain Monte Carlo (MCMC) simulations. using the Metropolis-Hastings algorithm at varying temperatures [Metropolis and Ulam, 1949]. Specifically, we simulate the 2D Ising model using the Metropolis-Hastings algorithm at varying temperatures, where each update involves independently flipping a single spin [Metropolis et al., 1953]. If the update decreases the overall energy, the spin flip is accepted. If the energy increases by ΔE , the spin flip is accepted with a probability given by

$$p = 1 - \exp(-\Delta E/kT). \quad (3.1)$$

Spins are updated in a typewriter ordering, ensuring that all spins are sequentially updated, thus mimicking the sequential updating by the injection pulses in the CIM [Böhm et al., 2018].

Figure 4 demonstrates snapshots of the spin configuration at $Z = 1,000$; 5,000; 100,000, and 500,000 as it approaches the ground state obtained by Monte Carlo simulations at different temperatures $T = \{0.01, 2.0, 2.5\}$. Starting from a random spin state, low-temperature Monte Carlo simulations exhibit the same clustering process as CIMs: smaller domains evaporate or merge to form larger

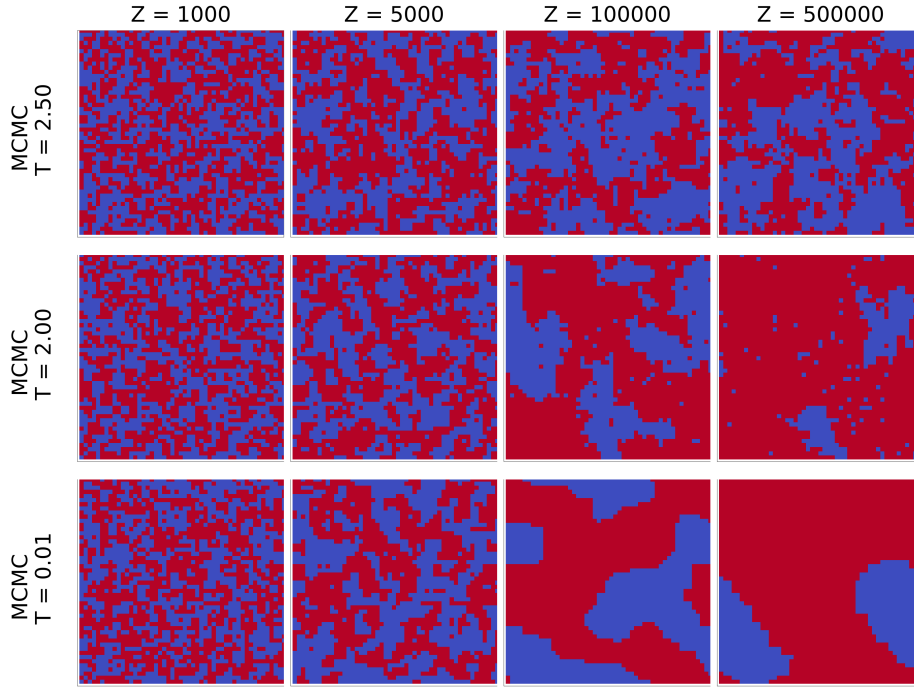


Figure 4: Snapshots of the spin configuration as it approaches the ground state obtained by Monte Carlo simulations (MCMC) at different temperatures $T = \{0.01, 2.0, 2.5\}$. *Parameters:* Step Size: $\Delta t = 0.05$; Number of Steps: 11,000.

ones until the ground state is reached. At higher temperatures, temperature-induced spin flips create irregularly shaped, temporary domains. As the phase transition of the square lattice model approaches, these domains grow, increasing correlation length. As such, despite operating at room temperature, we can conclude, in alignment with [Böhm et al., 2018], that CIMs best mimic low-temperature Ising model behavior. This suitability for simulating low-temperature spin systems is crucial for solving optimization problems.

3.2 Amplitude Heterogeneity

A significant performance challenge in classical CIMs is accurately mapping the Ising Hamiltonian onto the network’s loss landscape, a difficulty made worse by varying amplitudes among OPOs [Wang et al., 2013]. This heterogeneity is particularly problematic in frustrated spin systems. Amplitude heterogeneity occurs when OPO amplitudes deviate from their ideal values of $\pm\sqrt{\alpha}$, where α is the target pump parameter that is incrementally increased during the operation of the device [Leleu et al., 2019]. These deviations cause CIMs to frequently miss the true ground states of the Ising Hamiltonian, instead converging to the lowest energy eigenstate of the coupling matrix. From an optimization perspective, amplitude heterogeneity parallels the issue of infeasibility encountered when a constraint is replaced with a penalty function [Beloborodov et al., 2020].

3.2.1 Nonlinear Feedback Control

We can partially address the issue of amplitude heterogeneity by incorporating a nonlinear filter function such as $\tanh(\sum_j \xi J_{ij} x_j)$ for the feedback pulse. This approach, CIM-NLF, aims to achieve homogeneous OPO amplitudes and satisfy the necessary mapping conditions towards the system’s endpoint, especially well above threshold. However, nonlinear filtering alone is insufficient to prevent the system from becoming trapped in numerous local minima. As the problem size N increases, NP-hard Ising problems exhibit an exponentially increasing number of local minima, making the system highly susceptible to ineffective trapping [Reifenstein et al., 2021].

3.2.2 Chaotic Amplitude/Feedback Control

Alternatively, we can destabilize trapping sets of analog states corresponding to local minima of the binary spin Hamiltonian by expanding the phase space to incorporate error signals [Sourlas, 2005]. These signals correct amplitude heterogeneity in analog spin states while managing their velocity divergence. Specifically, in such systems, the feedback mechanism is adjusted to include an auxiliary variable (error pulse amplitude) $e_i > 0$, which acts as a non-Markovian reservoir ensuring positive entropy production despite dissipation [Leleu et al., 2019]. The spin variable x_i and auxiliary variable e_i follow these deterministic equations:

$$\frac{dx_i}{dt} = -x_i^3 + (p-1)x_i + e_i z_i \quad (3.2)$$

$$\frac{de_i}{dt} = -\beta(x_i^2 - \alpha)e_i. \quad (3.3)$$

Here, $\alpha > 0$ and $\beta > 0$ represent the target amplitude baseline and the timescale relative to the evolution of error variables, respectively. These parameters can be dynamically adjusted to optimize system performance over time. This integrated system of deterministic differential equations is referred to as CIM with chaotic amplitude control (CIM-CAC), owing to its inherent chaotic dynamics when applied in digital algorithms or optical CIMs mimicking these equations [?]. While destabilizing local minima inevitably makes the ground state unstable as well, allowing the machine to explore many local minima enables the identification of the lowest energy state subsequently. Alternatively, carefully modulating the system parameters yields a high probability of remaining in a ground state towards the end of the trajectory.

To gain deeper insights into the ways in which CIMs operate, in addition to CIM-CAC, we introduce an alternative modification that adheres to the same principles, known as CIM with chaotic feedback control (CIM-CFC):

$$\frac{dx_i}{dt} = -x_i^3 + (p-1)x_i + e_i I_i, \quad (3.4)$$

$$\frac{de_i}{dt} = -\beta((e_i I_i)^2 - \alpha)e_i, \quad (3.5)$$

The key difference between CIM-CAC and CIM-CFC is that in CIM-CFC, the time evolution of the error variable e_i monitors the feedback signal z_i rather than the internal amplitude x_i . Additionally, CIM-CFC exhibits slightly simpler dynamics, facilitating its numerical integration [Reifenstein et al., 2021].

Fixed Point Analysis To understand the similarities between CIM-CAC and CIM-CFC, we analyze their fixed points by imposing:

$$\frac{dx_i}{dt} = 0, \quad \frac{de_i}{dt} = 0, \quad \forall i. \quad (3.6)$$

For CIM-CAC, setting the time derivative of e_i to zero gives $x_i^2 = \alpha$, as both β and e_i are strictly positive. Moreover, imposing the equality defined in Eq. (3.10) onto Eq. (3.4) results in:

$$e_i = \frac{x_i^3 - (p-1)x_i}{\xi\{\mathbf{J}\vec{x}\}_i}. \quad (3.7)$$

At steady state, where the analog x_i 's are exactly binary with $x_i = \sigma_i \sqrt{a}$, this simplifies to:

$$e_i = \frac{1 - \alpha - p}{\xi h_i \sigma_i} \implies |e_i| \propto \frac{1}{|h_i|}, \quad (3.8)$$

where h_i are the elements of the vector $\vec{h} = \mathbf{J}\vec{\sigma}$. Moreover, we find that CIM-CFC shows nearly identical fixed points [Reifenstein et al., 2021], suggesting that despite the differences between CIM-CAC and CIM-CFC, their dynamics are similar.

Limit Cycles and Chaotic Attractors A critical condition for the existence of chaotic attractors is that the divergence of the velocity, or equivalently, the rate of change of phase space volumes, is negative near the trapping sets [Strogatz, 2018]. To examine the existence of limit cycles and chaotic attractors within the CIM-CAC/CFC algorithm, we follow [Leleu et al., 2019] and begin by rewriting the time evolution of the proposed CIM with chaotic amplitude control to be of the following equation:

$$\frac{dx_i}{dt} = (-1 + p)x_i - x_i^3 + \epsilon e_i \sum_j J_{ij} x_j, \quad (3.9)$$

where $e_i(t)$ is given by

$$e_i(t) = e_i(0) e^{-\beta \int_0^t (x_i(t')^2 - a(t')) dt'} \quad (3.10)$$

after integrating the time evolution of the auxiliary error variables. To estimate the divergence at configuration σ , we utilize the fact that the fast subsystem on $\mathbf{x}$ converges to the slow manifold characterized by $dx_i/dt \approx 0$, $\forall i$, in the limit $\beta \ll 1$, which can be approximated using the Taylor series expansion:

$$x_i = x_i^{(0)} + \epsilon x_i^{(1)} + \mathcal{O}(\epsilon^2) \quad (3.11)$$

for $p > 1$. At the zeroth order, the state is described as:

$$(x_i^{(0)})^2 = (-1 + p), \quad (3.12)$$

$$e_i^{(0)} = e_i(0) e^{-\beta \int_0^t (-1 + p - a(t')) dt'}. \quad (3.13)$$

Since $e_i^{(0)}$ is independent of the index i when $e_i(0) = 1$, $\forall i$, we can rewrite $e_i^{(0)}$ as $e^{(0)}$, which is the zeroth order approximation of $e_i(t)$. For $e_i(0) = 1$, $\forall i$, $e^{(0)}(t)$ is given by:

$$e^{(0)}(t) = e^{-\beta \int^t (-1 + p - a) dt'}. \quad (3.14)$$

At the first order, the state is described as:

$$x_i^{(1)} = \frac{e^{(0)} \sum_j J_{ij} x_j^{(0)}}{2(x_i^{(0)})^2} \quad (3.15)$$

$$e_i^{(1)} = -\beta e^{(0)} \int_0^t 2x_i^{(0)}(t') x_i^{(1)}(t') dt'. \quad (3.16)$$

The first order term $e_i^{(1)}$ depends on the previous states visited. The divergence of the state-space velocities $\mathbf{f}$ and $\mathbf{g}$ are given by

$$\text{div } \mathbf{f} = \sum_i \frac{\partial f_i}{\partial x_i} = N(1 - p) - 3 \sum_i x_i^2, \quad (3.17)$$

$$\text{div } \mathbf{g} = \sum_i \frac{\partial g_i}{\partial x_i} = -\beta \left(\sum_i x_i^2 - aN \right). \quad (3.18)$$

Using the Taylor approximation $x_i^2 \approx (x_i^{(0)})^2 + 2\epsilon x_i^{(0)} x_i^{(1)} + \mathcal{O}(\epsilon^2)$ in the limit of small coupling strength ($\epsilon \ll 1$), and noting that the states become approximately binary in this limit, the divergence of the vector fields $\mathbf{f}$ and $\mathbf{g}$ are approximated as:

$$\text{div } \mathbf{f} = -2N(-1 + p) + 6\epsilon e^{(0)}(t) \mathcal{H} + \mathcal{O}\left(\frac{(e^{(0)})^2 \epsilon^2}{p - 1}\right), \quad (3.19)$$

$$\text{div } \mathbf{g} = \beta \left[N(1 - p + a) + 2\epsilon e^{(0)}(t) \mathcal{H}(t) + \mathcal{O}\left(\frac{(e^{(0)})^2 \epsilon^2}{p - 1}\right) \right]. \quad (3.20)$$

Although this Taylor approximation is valid only for $p > 1$, numerical simulations show that it remains accurate by continuity for $p < 1$ and $|1 - p| \ll 1$ for a finite value of ϵ .

To prevent the system from becoming trapped in limit cycles and chaotic attractors, [Leleu et al., 2019] propose ensuring that the divergence of velocity in the auxiliary subspace e remains positive along the system trajectories. This can be achieved by modulating the target amplitude a as follows:

$$a = \alpha + \epsilon \langle e_i(t_c) h_i(t) \sigma_i(t) \rangle, \quad (3.21)$$

where α is the target amplitude baseline, t_c is the time of the last configuration change σ , and $\langle X_i \rangle$ denotes the ensemble average of X_i . This control scheme ensures that the divergence is approximately given by:

$$\text{div } \mathbf{g} \approx \kappa, \quad (3.22)$$

with $\kappa = N\beta(1 - p + \alpha)$. By modulating α and p such that the conditions for positive divergence are respected, the system avoids stable fixed points at suboptimal configurations and ensures positive divergence in the auxiliary subspace. When β is very small, the fast variables x are slaved to the slow error signal e . Thus, positive divergence in the auxiliary subspace implies that the entire system cannot be trapped in a region of negative divergence near an attractor.

Hardware Implementation [Reifenstein et al., 2021] designed a CIM-CAC and CIM-CFC system using optical error correction circuits with a single active device: a noise-free phase-sensitive amplifier (PSA). The other components are passive, allowing for potential on-chip integration and low-energy consumption. The system employs a main ring cavity storing signal pulses with normalized amplitudes x_i and error pulses with normalized amplitudes e_i for $i = 1, 2, \dots, N$. Signal pulses start from vacuum states $|0\rangle_1, |0\rangle_2, \dots, |0\rangle_N$ and are amplified or deamplified along the X -coordinate by a pump rate p . Error pulses begin from coherent states $|\alpha\rangle_1, |\alpha\rangle_2, \dots, |\alpha\rangle_N$ and are amplified by another pump rate p' .

By maintaining the squared amplitudes of the error pulses below 1 ($e_i^2 < 1$), the error pulses remain in a linear amplification/deamplification regime. Signal pulses operate in both linear amplifier/deamplifier ($x_i^2 < 1$) and nonlinear oscillator ($x_i^2 > 1$) regimes. An extraction beamsplitter (BS_e) selects parts of these pulses, which are then amplified to classical levels by a noise-free phase-sensitive amplifier (PSA_0). Although the extracted amplitudes $\tilde{x}_i$ and $\tilde{e}_i$ experience signal-to-noise ratio (SNR) degradation due to vacuum noise at BS_e , PSA_0 amplifies them sufficiently to prevent further SNR degradation despite large linear losses in the optical error correction circuits.

A fraction of the PSA_0 output is directed to an optical homodyne detector to measure the extracted signal and error pulses. Detection errors are influenced by the reflectivity of BS_e and the incident vacuum fluctuation. For instance, a signal pulse ($\tilde{x}_j$) enters PSA_j and then an optical delay line DL_j with a delay of $(2N - 2j + 1)\tau$. The phase-sensitive gain/loss of PSA_j is adjusted to $pG_j = \xi J_{ij}$, so the signal pulse reaching the fan-in circuit at $t = 2N\tau$ equals $\epsilon J_{ij} \tilde{x}_j$. Consequently, the fan-in circuit outputs a pulse with amplitude $\sum_j \xi J_{ij} \tilde{x}_j$. A 10 dB phase-sensitive gain/loss in PSA_j allows for an Ising coupling rate of $10^{-2} < |\xi J_{ij}| < 1$.

The fan-in circuit output is further amplified by another phase-sensitive amplifier PSA_e by a factor of $\sqrt{G_e} = \tilde{e}_i$, achieved by modulating PSA_e 's pump power based on the $\tilde{e}_i$ measurement. This output is reinjected into the signal pulse (x_i) of the main cavity via BS_i . The extraction beamsplitter BS_e outputs both signal and error pulses, but the latter are used only for homodyne detection. The pump power to PSA_0 is switched off for the error pulses to deamplify the residual error pulses by $\text{PSA}_1, \text{PSA}_2, \dots, \text{PSA}_N, \text{PSA}_e$, and prevent their spurious reinjection into the main cavity. The error pulse dynamics are then governed solely by the pump power p_i to the main cavity PSA , set to satisfy $p_i - 1 = \beta(\alpha - \tilde{x}_i^2)$ or $p_i - 1 = \beta(\alpha - \tilde{z}_i^2)$.

3.2.3 Separable Feedback Control

The final system we consider was developed by [Reifenstein et al., 2021] and features distinct equations governing its dynamics:

$$\frac{dx_i}{dt} = -x_i^3 + (p-1)x_i - f(cz_i) - k(z_i - e_i), \quad (3.23)$$

$$\frac{de_i}{dt} = -\beta(e_i - z_i). \quad (3.24)$$

Here, the non-linear function f is a sigmoid-like function, such as $f(z) = \tanh(z)$, and p, k, c , and β are system parameters. This model is called CIM with separated feedback control (CIM-SFC) because it explicitly separates the two essential aspects of CIM-CAC and CIM-CFC: amplitude heterogeneity correction and local minima destabilization. The term $f(cz_i)$ generates the mutual coupling and passively addresses amplitude heterogeneity, while the error signal e_i , introduced through the term $k(z_i - e_i)$, helps destabilise local minima. In doing so, we add another N degrees of freedom, which allows the machine to visit a local minimum, escape from it, and continue to explore nearby states. This is not possible with a conventional CIM algorithm.

A notable difference and key innovation in CIM-SFC is the linear differential equation for the error signal e_i concerning the mutual coupling signal z_i . Unlike CIM-CAC and CIM-CFC, the only non-linear components in CIM-SFC are the gain saturation term $-x_i^3$ and the function f . Additionally, the interpretation of the auxiliary variable e_i differs: in CIM-CAC and CIM-CFC, e_i is a strictly positive value that modulates the mutual coupling term exponentially. In contrast, in CIM-SFC, e_i stores sign information and spans the same range of values as the mutual coupling signal z_i . Consequently, the error signal e_i and the term $k(z_i - e_i)$ function as low-pass and high-pass filters, respectively, with $k(z_i - e_i)$ primarily detecting sharp changes in z_i .

Fixed Point Analysis In CIM-SFC, the fixed points are generally very complicated and difficult to express explicitly. By considering the limit of $c \gg 1$, however, the fixed points will take the form $x_i = \lambda\sigma_i$ and $e_i = \lambda h_i$, where λ is a number such that $-\lambda^3 + (p-1)\lambda = -1$. This formula is only valid if $f(cI)$ is an odd function that takes the value of $+1$ for $c \gg 1$ and -1 for $cz \ll -1$. Therefore, it is important to choose an appropriate function f . As such, while in CIM-CAC and CIM-CFC, the error signal is

$$|e_i| \propto \frac{1}{|h_i|}, \quad (3.25)$$

whereas in CIM-SFC, the error signal is

$$|e_i| \propto |h_i|. \quad (3.26)$$

This difference makes CIM-SFC more robust to quantum noise from reservoirs and pump sources.

Hardware Design Implementation CIM-SFC employs a design similar to that of CIM-CAC and CIM-CFC. The feedback signal $\tilde{z}_i = \sum_j J_{ij}\tilde{x}_j$ is deamplified by PSA_e with an attenuation coefficient $\tanh\left(\frac{\tilde{z}_i^{(m)}}{\tilde{z}_i^{(m)}}\right)$, where $\tilde{z}_i^{(m)}$ is the optical homodyne measurement result of $\tilde{z}_i$. This feedback signal is then re-injected into the signal pulse x_i in the main cavity through BS_i .

Part of the fan-in circuit output $\tilde{z}_i$ is delayed by a delay line DL_e with a delay time of $N\tau$ and combined with the error pulse e_i inside the main cavity. This process, governed by PSA_e , deamplifies the error signal and implements the error correction signal amplitude $e_i - \tilde{z}_i$, which is then coupled to the signal pulse x_i within the main cavity using a standard optical delay line. In addition, the term $-\beta(e_i - \tilde{z}_i)$ is implemented by a phase-sensitive amplifier PSA_e of the main cavity.

4 Numerical Simulation of Coherent Ising Machine Variations

In this section, we delve into the dynamics of the error correction schemes proposed in this paper: CIM-CAC, CIM-CFC, and CIM-SFC. Despite being governed by distinct equations, these systems share a common conceptual foundation. We will examine their similarities, correlation properties, and differences in their trajectories.

4.1 Similarity Analysis

To understand the similarities between CIM-CAC/CIM-CFC and CIM-SFC, we will consider the system described below. We introduce the 'mutual-coupling signal' $M_i(t) = \sum_j \xi J_{ij} x_j(t)$ and the 'injection feedback signal' $I_i(t)$. Both CIM-CAC/CIM-CFC and CIM-SFC can be expressed as:

$$\frac{dx_i}{dt} = -x_i^3 + (p-1)x_i - I_i(t), \quad (4.1)$$

where $I_i(t)$ depends on the time evolution of $M_i(t)$. Figure 5 shows how $I_i(t)$ varies with respect to the mutual coupling field $M_i(t)$ for four different feedback schemes. Specifically:

- For CIM, $I(t) = M(t)$.
- For CIM-NLF, $I(t) = \tanh(M(t))$.
- For CIM-CFC, $I(t) = e(t)M(t)$, where $de(t)/dt = -\beta e(t) (e(t)^2 M(t)^2 - \alpha)$.
- For CIM-SFC, $I(t) = \tanh(cM(t)) + k(M(t) - e(t))$ where $de(t)/dt = -\beta(e(t) - M(t))$.

The similarity between CIM-CAC/CIM-CFC and CIM-SFC is as follows: if the mutual coupling field $M_i(t)$ retains the constant for a certain period of time, then the injection feedback field $I_i(t)$ will converge on the value given by $\text{sign}(M_i(t))$. However, if $M_i(t)$ varies sharply, then $I_i(t)$ will deviate from its steady state values: $+1/-1$. This small deviation is effective for triggering destabilization when the system is near a local minimum, which allows the machine to explore new spin configurations.

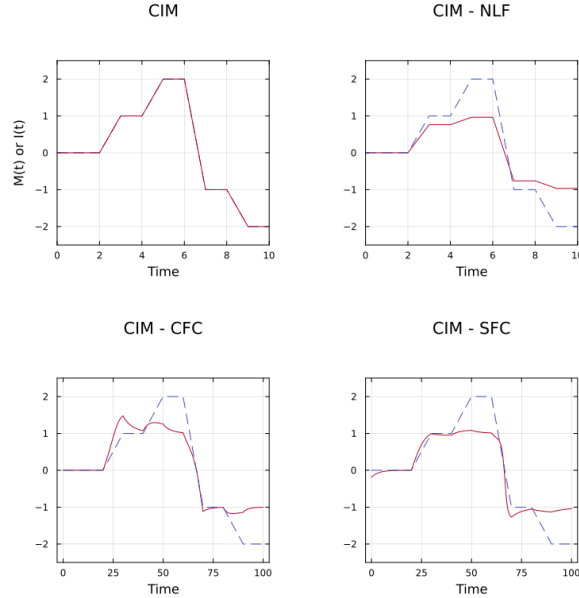


Figure 5: Mutual coupling field (blue/dashed) and injection feedback field (red/solid) in four different feedback systems.

4.2 Correlation Properties

A defining characteristic of classical Coherent Ising Machines (CIM) and CIM with Nonlinear Feedback (NLF) is their deterministic nature, where closely related initial conditions yield nearly identical trajectories. Figure 6 demonstrates this property by depicting the correlation of pulse amplitudes between two closely related initial conditions. Specifically, the initial pulse amplitude x_1 (plotted on the x-axis) is drawn from a zero-mean Gaussian distribution with a standard deviation of 0.25. The second initial condition, x_2 (plotted on the y-axis), is derived from x_1 with a small amount of added noise (standard deviation of 0.001). The correlation remains consistent even after $N = 1000$ steps, indicating robust deterministic behavior over time.

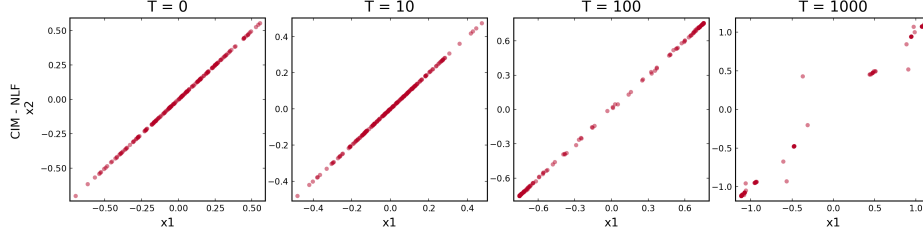


Figure 6: Signal pulse amplitude correlations at different evolution time in CIM-NLF.

Consequently, we analyze the presence of this deterministic property across three algorithms designed to mitigate amplitude heterogeneity and escape local minima. Although CIM-CAC, CIM-CFC, and CIM-SFC are grounded in similar principles, their dynamics exhibit notable distinctions. CIM-CAC and CIM-CFC frequently demonstrate chaotic behavior, characterized by trajectories highly sensitive to initial conditions [Leleu et al., 2019, Kako et al., 2020].

In Figure 7, we observe that despite closely matched initial conditions, CIM-CAC and CIM-CFC trajectories diverge after approximately 100 steps, underscoring their sensitivity to initial perturbations. This sensitivity persists across varying parameters and initial conditions. Conversely, CIM-SFC typically converges to stable periodic orbits, maintaining the initial correlation even after $N = 1000$ steps. While CIM-SFC generally retains deterministic behavior, deviations can occur under specific parameter settings or initial conditions, particularly when parameters are dynamically adjusted. This nuanced behavior suggests that while CIM-SFC is less sensitive to initial conditions compared to CIM-CAC and CIM-CFC, occasional chaotic dynamics may arise during optimization tasks, especially under parameter modulation [Reifenstein et al., 2021].

4.3 Differences in Trajectories

To qualitatively observe the differences in system dynamics, we examine the trajectories of each system. Figure 8 presents trajectory examples for all three systems applied to a Sherrington-Kirkpatrick (SK) spin glass problem with a size of $N = 200$, featuring both fixed and linearly modulated parameters. In this instance, the coupling matrix exhibits perfect antiferromagnetic coupling, where nearest neighbour couplings are set to -1 and all others are zero.

When parameters are fixed, distinct behaviours emerge among the three systems: CIM-SFC and CIM-CAC rapidly become trapped in a stable periodic attractor (Figure 8, bottom-left), whereas CIM-CAC continue to explore the state space unpredictably. This necessitates the slow modulation of parameters in CIM-SFC to enable the system to locate the ground state. While CIM-CFC and CIM-CAC can achieve ground states with fixed parameters (e.g., Figure 8, top-right), parameter modulation significantly enhances their performance.

Moreover, parameter modulation over time reduces the likelihood of the systems becoming trapped in local minima and exhibiting oscillatory behavior, where the state perpetually transitions between different states without escaping. This modulation facilitates a more effective search for the ground state, improving overall system performance.

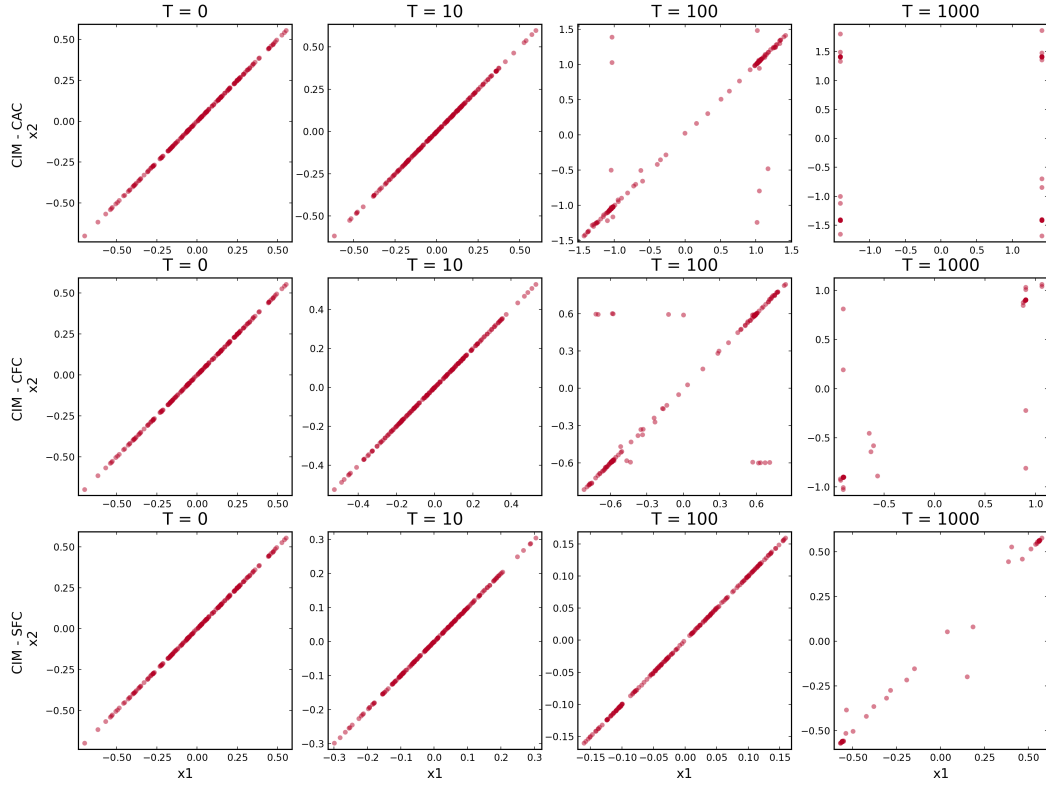


Figure 7: Signal pulse amplitude correlations at different evolution time in CIM-CAC, CIM-CFC and CIM-SFC.

In the bottom right panel of Figure 8, the parameters c and p of CIM-SFC are linearly increased (with p ranging from -1 to +1 and c from 1 to 3). As these parameters change, the system transitions between attractors, ultimately settling into a fixed point or local minimum. This gradual increase in c and p transitions the nonlinear term $\tanh(cz_i)$ from a "soft spin" mode, where it varies continuously between -1 and 1, to a "discrete" mode, where $\tanh(cz_i)$ predominantly takes on values of +1 or -1. This transition is essential for the proper functioning of CIM-SFC [Reifenstein et al., 2021].

5 Energy Reduction in Coherent Ising Machine Systems

This section discusses ways of reducing energy in coherent Ising machine systems by implementing a pump pulse scheme utilising second harmonic generation (SHG) pulses. This approach supplies the main cavity PSA, post amplifier PSA_0 , delay line amplifiers PSA_1 , PSA_2 , ..., PSA_N , and exit amplifier PSA_e , thereby minimizing reliance on energy-intensive EOM modulators.

A soliton frequency comb generator produces a pulse train with a repetition frequency of 100 GHz and a wavelength of 1.56 μm . This pulse train is initially amplified by a pump amplifier PSA_p before being split into multiple branches. N storage ring cavities continuously generate pump pulses for PSA_1 , PSA_2 , ..., PSA_N , facilitating the matrix-vector multiplication (MVM) $\sum \xi J_{ij} \tilde{x}_j$. The pulses stored in the i th ring cavity are amplified to appropriate amplitudes to achieve a gain of $\sqrt{G_{ij}} = \xi J_{ij}$. This process allows the use of N EOM arrays in just one round trip of the ring cavity, totaling a required time of $N \times 10$ ps.

Out-coupling losses in the storage ring cavities are compensated by the linear gain provided by the internal PSA_s . The pump pulses for PSA_p , PSA_s , and PSA_0 maintain constant amplitudes and are directly driven by the PSA_p . In contrast, the pump pulses P and P_i for the signal and error pulses in the main cavity, as well as the exit PSA_e , require modulation throughout the computation period.

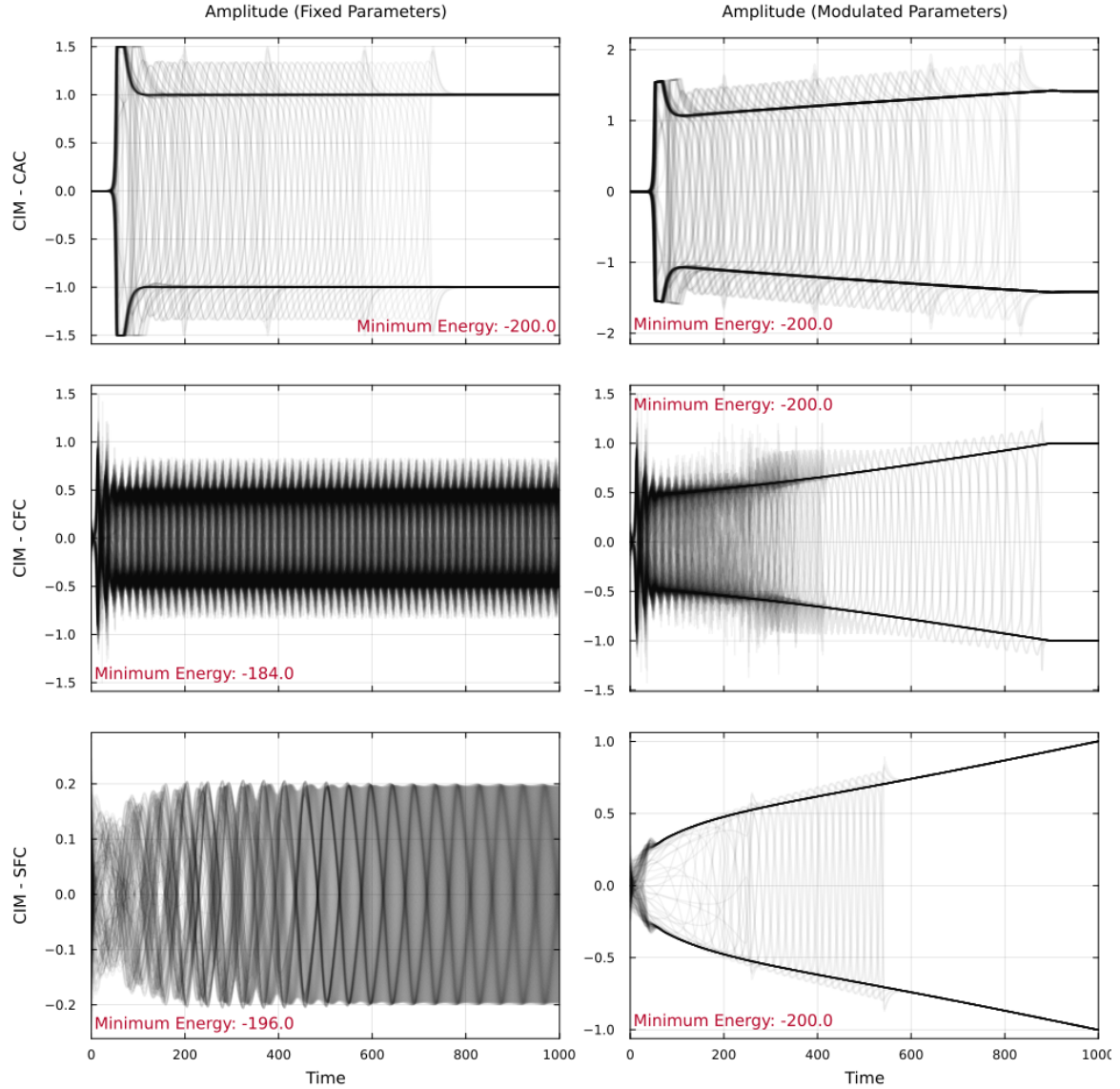


Figure 8: Signal pulse amplitude trajectories of CIM-CAC, CIM-CFC, and CIM-SFC with fixed and modulated parameters.

(i) For CIM-CAC, we restrict the x_i variables to the range $[-1.5\sqrt{\alpha}, 1.5\sqrt{\alpha}]$ at each time step. The initial value x_i is set to a random value chosen from a zero-mean Gaussian distribution with a standard deviation of 10^{-4} and $e_i = 1$. *Parameters:* (fixed) $p = -1, \alpha = 1, \beta = 0.2$. Step size: $\Delta t = 0.125$. (modulated) $p = -1 \rightarrow 1$ over 900 time steps, then $p = 1$ for 100 time steps; $\alpha = 1 \rightarrow 2$ for 900 time steps, then $\alpha = 2$ for 100 time steps; $\beta = 0.2$. Step size: $\Delta t = 0.125$.

(ii) For CIM-CFC, we restrict the x_i variables to the range $[-1.5, 1.5]$ and the e_i variables to the range $[0.01, \infty]$. The initial value x_i is set to a random value chosen from a zero-mean Gaussian distribution with a standard deviation of 0.1 and $e_i = 1$. *Parameters:* (fixed) $p = -1, \alpha = 1, \beta = 0.2$. Step size: $\Delta t = 0.4$. (modulated) $p = -1 \rightarrow 1$ over 900 time steps, then $p = 1$ for 100 time steps; $\alpha = 1, \beta = 0.2$. Step size: $\Delta t = 0.4$.

(iii) For CIM-SFC, we do not restrict the x_i and e_i variables, as the system is numerically stable. The initial value x_i is set to a random value chosen from a zero-mean Gaussian distribution with a standard deviation of 0.1 and $e_i = 0$. *Parameters:* (fixed) $p = -1, c = 1, \beta = 0.3, k = 0.2$. Step size: $\Delta t = 0.4$. (modulated) $p = -1 \rightarrow 1, c = 1 \rightarrow 3, \beta = 0.3 \rightarrow 0.1, k = 0.2$. Step size: $\Delta t = 0.4$.

For all simulations, $\xi = \sqrt{2N / \sum J_{ij}^2}$.

Additionally, in simulation studies, the Ising energy was calculated at each time step (round trip), with the smallest energy obtained used as the computation result. Translating this approach to an optical implementation would require measuring the $\tilde{x}_i$ amplitude at every round trip and calculating the energy using an external ADC/FPGA circuit. Since this approach would introduce delays and increase energy consumption, thereby negating the advantages of using optics, an alternative is sought. [Reifenstein et al., 2021] found that by applying proper parameter modulation, it is sufficient to use only the final state of the system to achieve a high probability of success. Specifically, for an 800-spin Ising instance (SK model), if CIM-SFC visits the ground state at any point during its trajectory, it will be in the ground state at the end of the trajectory 100% of the time. In contrast, for CIM-CFC and CIM-CAC, this was true only 75% and 48% of the time, respectively. The authors suggest that this difference arises from both the intrinsic dynamics and the parameters used.

Therefore, for a CIM with optical error correction, it is possible to digitize only the final measured value of $\tilde{x}_i$ after many round trips to obtain a computational result, maintaining a high probability of success. For CIM-CFC and CIM-CAC, it may be beneficial to read the spin configuration multiple times during the last few round trips, as these machines often visit the ground state near the end of the trajectory, even if they do not remain there [Reifenstein et al., 2021].

6 Experiments

To provide a clearer comparison of the three algorithms CIM-CAC, CIM-CFC, and CIM-SFC, we will assess their efficiency in solving the NP-hard combinatorial optimization problem, MAX-CUT. The MAX-CUT problem, given an undirected graph $G = (V, E)$ with an edge weight function $w : E \rightarrow \mathbb{R}$ (where E and V denote the edges and vertices, respectively), aims to find a cut $(S, V \setminus S)$ that maximizes the sum of the weights of the edges crossing the cut over all possible cuts [Karp, 1972].

Concretely, let $w_{jl} = w_{lj}$ be the edge weight if $(j, l) \in E$ and $w_{jl} = 0$ if $(j, l) \notin E$. Let $\sigma_j = +1$ if the j -th vertex is in S and $\sigma_j = -1$ otherwise. The weight of a cut can be expressed as:

$$w(S) = \sum_{j \in S, l \in V \setminus S} w_{jl} = \frac{1}{4} \sum_{j, l \in V} w_{jl} - \frac{1}{4} \sum_{j, l \in V} w_{jl} \sigma_j \sigma_l. \quad (6.1)$$

When the coupling coefficient J_{jl} in the Ising model is set to $-w_{jl}$, a maximum cut of the given graph corresponds to a ground state of the Ising problem, and vice versa [Wang et al., 2013].

We will use the G-Set instances as benchmarks for MAX-CUT performance¹. Specifically, we will consider 6 instances of size 800, covering three different graph structures: (i) random graphs, (ii) toroidal graphs, and (iii) planar graphs. These instances have varying edge densities and include edge weights of either $+1$ and 0 , or $+1$, 0 , and -1 . We employ a

6.1 Parameter Choice

We can approximate the average residual energy visited by CIM-CAC/CIM-CFC during the search process using the formula:

$$\Delta E_{\text{avg}} \approx K \frac{1-p}{\alpha\beta}, \quad (6.2)$$

where K is a constant that depends on the problem type and size. This allows us to predict the effective sampling temperature of the system, even though the distribution may not follow an exact Boltzmann distribution [Leleu et al., 2019].

Based on this principle, we gradually reduce the “system temperature” to achieve an annealing effect, which can be accomplished by increasing p and α . By selecting different ranges for p for various G-Set instances, we can account for the significantly different values of the constant K that depend on the structure of the MAX-CUT problem. We note here that, in a broader context, the

¹The G-Set instances are available at <https://web.stanford.edu/~yyye/yyye/Gset/>.

value of K can be predicted based on the problem type, allowing us to choose appropriate ranges for p and α .

CIM - CAC Following the approach in [Reifenstein et al., 2021], we imposed a constraint on the x_i variables for the CIM-CAC algorithm, limiting them to the range $[-1.5\sqrt{\alpha}, 1.5\sqrt{\alpha}]$ at each time step. The algorithm parameters are linearly adjusted from their initial to final values over T_r time steps and then maintained at the final values for an additional T_p time steps. The initial values of x_i are randomly selected from a zero-mean Gaussian distribution with a standard deviation of 10^{-4} , and e_i is set to 1.

For all G-Set instances, we modulate α linearly from 1.0 to 3.0, while keeping β constant at 0.3. Furthermore, the parameters p , ΔT , and the number of time steps for each phase are selected based on the instance type, as detailed below:

Table 1: Parameters for solving MAX-CUT instances on the G-Sets using CIM-CAC.

Graph Type	N	Edge Weight	G-Set Instance	p	ΔT	T_r	T_p
Random	800	$\{+1\}$	1, 2	$-0.5 \rightarrow 1.0$	0.075	6000	666
Toroidal	800	$\{+1, -1\}$	11, 12	-4.0	0.1	4500	500
Planar	800	$\{+1\}$	14, 15	-1.0	0.05	18000	2000

CIM - CFC We imposed a constraint on the x_i variables for the CIM-CFC algorithm, limiting them to the range $[-1.5, 1.5]$ at each time step. Similarly, we restrict the e_i variables to the range $[0.01, \infty]$ at each time step. The algorithm parameters are linearly adjusted from their initial to final values over T_r time steps and then maintained at the final values for an additional T_p time steps. The initial values of x_i are randomly selected from a zero-mean Gaussian distribution with a standard deviation of 0.1 and e_i is set to 1.

For all G-Set instances, we keep α and β fixed, with respective values of 1.0 and 0.15. Furthermore, the parameters p , ΔT , and the number of time steps for each phase are selected based on the instance type, as detailed below:

Table 2: Parameters for solving MAX-CUT instances on the G-Sets using CIM-CFC.

Graph Type	N	Edge Weight	G-Set Instance	p	ΔT	T_r	T_p
Random	800	$\{+1\}$	1, 2	$-1.0 \rightarrow 1.0$	0.125	3600	400
Toroidal	800	$\{+1, -1\}$	11, 12	$-3.0 \rightarrow -1.0$	0.25	1800	200
Planar	800	$\{+1\}$	14, 15	$-2.0 \rightarrow 0.0$	0.125	7200	800

CIM-SFC We require the term $\tanh(cz_i)$ to transition from the 'soft-spin' mode, where $cz_i \approx 0$ and $\tanh(cz_i) \approx cz_i$, to the 'discrete-spin' mode, where $|cz_i| \gg 0$ and $\tanh(cz_i) \approx \sin(cz_i)$. To achieve this, we utilize the normalizing factor [Reifenstein et al., 2021]:

$$\xi = \sqrt{\frac{2N}{\sum J_{ij}^2}}, \quad (6.3)$$

which ensures that z_i will, on average, be around $\sqrt{2}$ for a randomly chosen spin configuration. This normalization allows us to use the same value for c in all cases and obtain consistent results. However, this approach is only effective for instances like SK instances, where each node has equal connectivity, leading to z_i having a similar range of values for all i .

Specifically, for CIM-SFC, no restriction on the values is needed as the system is more numerically stable. The parameters p , c , and β are modulated linearly from their initial values to their final

values during the simulation. The initial value of x_i is set to a random value chosen from a zero-mean Gaussian distribution with a standard deviation of 0.1, and e_i is set to 0. For all G-Set instances, p is modulated linearly from -1.0 to 1.0 . Additionally, we select the parameters $c, \beta, \Delta T$, and the number of time steps for each phase according to the instance type as follows: (Note here that 2000 node instances have not been tested.)

Table 3: Parameters for solving MAX-CUT instances on the G-Sets using CIM-CFC.

Graph Type	N	Edge Weight	G-Set Instance	c	β	k	ΔT	N Steps
Random	800	$\{+1\}$	1, 2	$1.0 \rightarrow 3.0$	$0.3 \rightarrow 0.0$	0.2	0.15	2666
Toroidal	800	$\{+1, -1\}$	11, 12	1.4	$0.05 \rightarrow 0.0$	0.32	0.4	2500
Planar	800	$\{+1\}$	14, 15	$-2.0 \rightarrow 0.0$	$0.05 \rightarrow 1.0$	0.2	0.125	5000

6.2 Results

Table 4 shows the best MAX-CUT value obtained using the previously mentioned parameters, in a run of 200 iterations. As demonstrated in our results, none of the methods managed to find the ground solution within the chosen number of time steps. This outcome does not align with the findings of [Leleu et al., 2019], which suggest that the coherent Ising machine with chaotic amplitude control can find low-energy states of the Ising Hamiltonian at least as efficiently as current state-of-the-art heuristics, even when simulated on a classical computer.

We hypothesize that this discrepancy may be due to errors in parameter modulation, indicating a need for larger grid searches, or simply that our choice of 200 iterations was insufficient. Although the results show promise, the configurations frequently became trapped in local minima despite high operational performance. Nonetheless, we observe that CIM-CAC outperforms CIM-SFC and CIM-CFC, consistently yielding the best solutions at each time point.

Table 4: Best MAX-CUT value obtained using CIM-CAC, CIM-CFC, and CIM-SFC.

ID	N	Graph Type	Edge Weight	C_{OPT}^*	C_{CAC}^*	C_{CFC}^*	C_{SFC}^*
1	800	Random	$\{+1\}$	11624	11114	10926	10730
2	800	Random	$\{+1\}$	11620	11042	11014	10732
11	800	Toroidal	$\{+1, -1\}$	564	440	440	432
12	800	Toroidal	$\{+1, -1\}$	556	438	440	424
14	800	Planar	$\{+1\}$	3064	2713	2704	2235
15	800	Planar	$\{+1\}$	3050	2657	2695	2330

7 Conclusion

This thesis has explored the challenges and potential solutions in improving the performance of Coherent Ising Machines (CIMs) for solving complex combinatorial optimization problems. We have examined in detail three key approaches to addressing the issues of amplitude heterogeneity and local minima trapping in CIMs: Chaotic Amplitude Control (CIM-CAC), Chaotic Feedback Control (CIM-CFC), and Separable Feedback Control (CIM-SFC).

Our analysis has revealed that while these methods share conceptual similarities, they exhibit distinct dynamics and performance characteristics. CIM-CAC and CIM-CFC demonstrate chaotic behavior and high sensitivity to initial conditions, potentially allowing for a more thorough exploration of the solution space. In contrast, CIM-SFC typically converges to stable periodic orbits, maintaining deterministic behavior under most conditions. Through numerical simulations on various G-Set instances of the MAX-CUT problem, we have evaluated the effectiveness of these approaches across different graph types and sizes. While our results did not conclusively demonstrate superiority of any

single method across all problem instances, they highlighted the potential of these enhanced CIM algorithms to tackle NP-hard optimization problems. The energy reduction strategies discussed, particularly the pump pulse scheme utilizing second harmonic generation, offer promising avenues for improving the efficiency of optical implementations of CIMs. These advancements could pave the way for more energy-efficient and scalable quantum-inspired computing systems.

Looking ahead, further research is needed to fully understand the parameter dependencies of these algorithms and to optimize their performance across a broader range of problem types and sizes. One promising avenue is the integration of the coherent Ising machine with machine learning. For instance, state-of-the-art reinforcement learning techniques have been used to train agents that can determine optimal parameters based on the specifics of the problem instance [Beloborodov et al., 2020]. Additionally, it would be valuable to explore whether incorporating prior information into the initial spin configuration could improve performance. Our findings suggest that the main drawback of the Ising machine, particularly when simulating, is its strong dependence on an optimal initial spin configuration.

Acknowledgments

I hereby declare that this report and the work described in it, submitted for assessment in the MPhil in Data Intensive Science, are my own, unless stated otherwise. I further declare no usage of ChatGPT throughout its completion, except for the generation of pseudocode. In such cases, the asked prompt was: "Generate LaTeX compatible PseudoCode: `relevant code`". I did utilise the autocomplete feature of GitHub Copilot for crafting comments and composing docstrings.

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